

Calculus I Lesson 10

1) Find $\lim_{x \rightarrow \infty} \frac{x-4}{x^2}$

Hint: $\frac{x-4}{x^2} = \frac{x/x^2 - 4/x^2}{x^2/x^2} = \frac{1/x - 4/x^2}{1}$

$$\lim_{x \rightarrow \infty} \frac{x-4}{x^2} = \underline{\hspace{2cm} ? \hspace{2cm}}$$

2) Find $\lim_{x \rightarrow \infty} \frac{3x-1}{x^2+3x+2}$

Hint: $\frac{3x-1}{x^2+3x+2} = \frac{3x/x^2 - 1/x^2}{x^2/x^2 + 3x/x^2 + 2/x^2} = \frac{3 - 1/x^2}{1 + 3/x + 2/x^2}$

$$\lim_{x \rightarrow \infty} \frac{3x-1}{x^2+3x+2} = \underline{\hspace{2cm} ? \hspace{2cm}}$$

3) Find $\lim_{x \rightarrow \infty} \frac{x^3 + 3x - 4}{x^2 + 2}$

Hint: $\frac{x^3 + 3x - 4}{x^2 + 2} = \frac{x^3/x^2 + 3x/x^2 - 4/x^2}{x^2/x^2 + 2/x^2} = \frac{x + 3/x - 4/x^2}{1 + 2/x^2}$

$$\lim_{x \rightarrow \infty} \frac{x^3 + 3x - 4}{x^2 + 2} = \underline{\hspace{2cm}} ?$$

4) Find $\lim_{x \rightarrow \infty} \frac{x^4 + 6x - 3}{x + 1}$

Hint: $\frac{x^4 + 6x - 3}{x + 1} = \frac{x^4/x + 6x/x - 3/x}{x/x + 1/x} = \frac{x^3 + 6 - 3/x}{1 + 1/x}$

$$\lim_{x \rightarrow \infty} \frac{x^4 + 6x - 3}{x + 1} = \underline{\hspace{2cm}} ?$$

5) Find $\lim_{x \rightarrow \infty} \frac{6x^2 - 5x + 1}{2x^2 + 8x + 2}$

Hint: $\frac{6x^2 - 5x + 1}{2x^2 + 8x + 2} = \frac{6x^2/x^2 - 5x/x^2 + 1/x^2}{2x^2/x^2 + 8x/x^2 + 2/x^2} = \frac{6 - 5x/x + 1/x^2}{2 + 8/x + 2/x^2}$

$$\lim_{x \rightarrow \infty} \frac{6x^2 - 5x + 1}{2x^2 + 8x + 2} = \underline{\hspace{2cm} ? \hspace{2cm}}$$

